

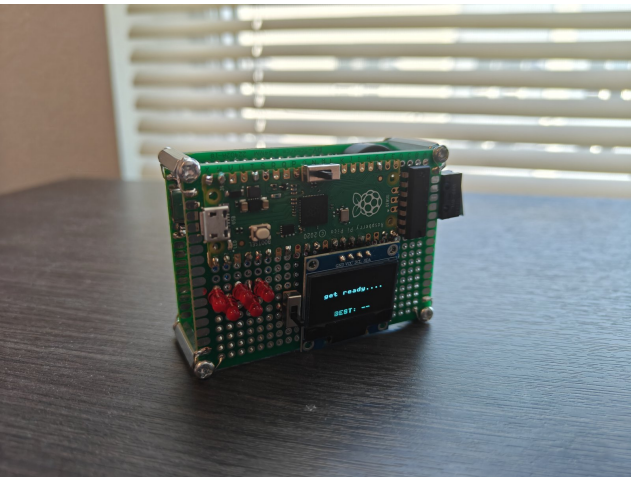
Reaction Time Game

A handheld reaction timer for Formula 1 fans that runs the F1 start-light sequence, with its timing verified on an oscilloscope

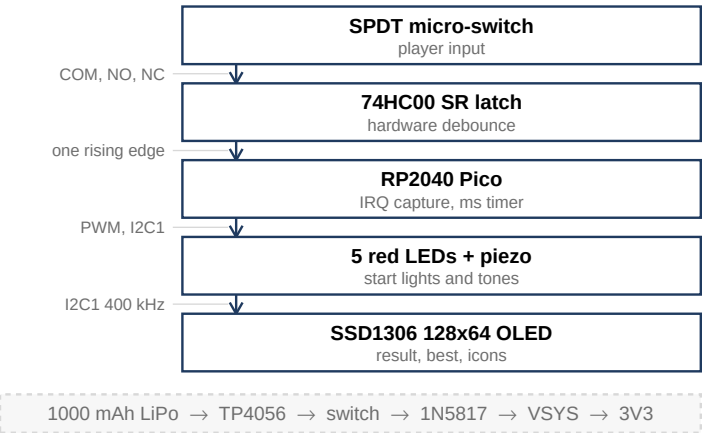
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Full documentation



SYSTEM ARCHITECTURE



WHAT IT DOES

Built for Formula 1 fans to test their reaction time against the start sequence drivers face: five red lights come on one second apart, then go out together after a random delay. A button press stops a millisecond timer and the result appears on an OLED screen. It runs on a Raspberry Pi Pico microcontroller from a rechargeable battery, and the button is debounced in hardware.

HOW IT WORKS

- SPDT micro-switch into a cross-coupled NAND latch: 6 to 11 transitions per press raw, one edge out.
- The Pico's RP2040 chip catches that edge on an interrupt armed only during a round; the same handler rejects a false start.
- 1000 mAh LiPo through a protected TP4056 and a Schottky diode-OR onto VSYS; switch measured at 0.2 mV off.

MEASURED RESULTS

My own reaction time, median of 30 rounds	202 ms
Timer error vs oscilloscope, mean of 10	+6.4 ms
Switch bounce edges, raw vs after latch	6–11 → 1
Supply voltage with power switch off	0.2 mV

KEY DECISIONS

- Hardware SR latch over RC debounce: an RC filter adds 1 to 3 ms to every timestamp.
- Charging with the load switched off keeps termination clean and validates the VSYS gauge.

MEASURED EVIDENCE



Fig. 1 — FNIRSI 2C53T, 0.5 ms/div, 1 V/div, DC: NO opens, NC closes 0.94 ms later



Fig. 2 — FNIRSI 2C53T, 50 ms/div, CH1 20 mV/div AC, CH2 500 mV/div DC, single shot